Airbnb Price Prediction Using Machine Learning

Peter Capriotti

***Abstract*—This project focuses on predicting Airbnb listing prices in Chicago using machine learning techniques. Accurate price estimation is important for both hosts seeking to optimize revenue and guests aiming to find fair market value. The dataset used was sourced from the Inside Airbnb Chicago dataset and includes approximately 1,766 cleaned listings filtered to exclude extreme outliers. Key features include property characteristics (bedrooms, bathrooms), geographic information (latitude and longitude), and user-generated signals (review scores and amenities).**

The methodology involved data cleaning, feature engineering, and model training. New features such as amenity counts and binary indicators for premium amenities (e.g., pool, gym, parking) were created. Two models were evaluated: a Random Forest Regressor as a baseline and a Gradient Boosting Regressor as the final model. Hyperparameters were tuned using grid search, and a log transformation was applied to address skewness in price distribution.

The final Gradient Boosting model achieved an R^2 score of 0.68 and a mean absolute error (MAE) of 44.1, significantly outperforming the initial Random Forest model (R^2 = 0.31). Results indicate that spatial features, capacity-related variables, and review metrics are strong predictors of price. Future improvements could include incorporating temporal trends and additional location-based features.

I. Introduction

Airbnb pricing is a complex problem influenced by multiple factors, including property characteristics, location, and user perception. Setting an optimal price is critical for hosts to maximize occupancy and revenue, while guests benefit from transparent and fair pricing. However, determining the right price manually can be difficult due to the nonlinear and interactive nature of these variables. This project aims to develop a machine learning model that can accurately predict nightly Airbnb prices in Chicago. The primary beneficiaries of this work include Airbnb hosts, who can use predictive insights to optimize pricing strategies, and platform analysts interested in understanding key drivers of price variation. The scope of this project includes data preprocessing, feature engineering, model selection, and evaluation using structured listing data. It focuses specifically on listings priced under 1000 to reduce the impact of extreme outliers.

The central research question is: How accurately can Airbnb listing prices be predicted using structured listing features and machine learning models, and which features contribute most to price variation?

II. Related Work

Airbnb price prediction has been widely studied using publicly available datasets such as the Inside Airbnb dataset. Researchers have found that listing price is influenced by a mix of property features location and user feedback such as reviews and ratings. Because of this complexity machine learning models are commonly used to improve prediction accuracy.

Earlier work often used linear regression to estimate Airbnb prices based on features like number of bedrooms bathrooms and location. While these models are easy to interpret they are limited because they cannot capture complex relationships between variables. For example the value of an amenity like a pool may depend on the neighborhood or type of listing which linear models do not handle well [1].

To address this issue many studies use ensemble models such as Random Forests. Random Forests combine many decision trees to improve prediction accuracy and reduce overfitting. These models can capture nonlinear relationships and interactions between features which makes them a strong baseline for price prediction tasks [2]. However they can still struggle with spatial data such as latitude and longitude especially if those features are not transformed or grouped into meaningful regions [3].

More advanced approaches use boosting methods such as Gradient Boosting and XGBoost. These models build trees sequentially where each new tree focuses on correcting the errors of the previous one. This allows boosting methods to better learn complex patterns in the data and often results in higher accuracy compared to Random Forest models [4]. Another important area in prior work is feature engineering. Studies show that creating new features such as total amenity count grouping locations into neighborhoods and using review based metrics can significantly improve model performance. In addition many researchers apply a log transformation to price because Airbnb prices are usually right skewed meaning there are many low priced listings and a few very expensive ones [1].

This project builds on previous research by combining feature engineering with ensemble learning methods. A Random Forest model is used as a starting point and a Gradient Boosting Regressor is used as the final model. Unlike simpler approaches this project includes engineered features such as amenity counts and grouped location data which helps the model better capture patterns in pricing.

III. Data Description

The dataset used in this project was obtained from the Inside Airbnb Chicago dataset, which provides publicly available information on Airbnb listings. The dataset contains detailed listing-level data including pricing, property characteristics, location, and user-generated review metrics. After cleaning and filtering, the final dataset consisted of approximately 1,766 listings.

The target variable for this study is the nightly price in USD. Listings with prices below 10 dollars or above 1000 dollars were removed to eliminate extreme outliers and ensure a more stable modeling range. Key input features include bedrooms, bathrooms, accommodates, review scores, latitude, longitude, and room type. These variables capture property size, geographic location, and perceived quality. The data was originally collected through web scraping by the Inside Airbnb project, which aggregates publicly available listing information from the Airbnb platform. Because the raw dataset contained inconsistencies such as missing values, misaligned columns, and formatting issues, significant preprocessing was required before analysis.

Exploratory data analysis was conducted to better understand the dataset. Summary statistics showed that price distribution was highly right skewed, with a large number of lower priced listings and a small number of high priced properties. Visualizations confirmed several important relationships. Price generally increased with the number of bedrooms and bathrooms, indicating that capacity is a strong driver of cost. Boxplots comparing room types showed that entire homes or apartments had significantly higher prices than private or shared rooms. Geographic patterns were also observed, with listings in central areas such as downtown Chicago tending to have higher prices.

These findings motivated the use of feature transformations and nonlinear models in later stages of the project.

IV. Methodology

The methodology for this project consists of data preprocessing, feature engineering, model training, and evaluation.

The dataset required substantial cleaning due to formatting inconsistencies. Price values were stored as strings with currency symbols and commas, which were removed and converted into numeric format. Rows with extreme or invalid values were filtered out, including listings with unrealistic bedroom or bathroom counts. Key numerical features such as bedrooms, bathrooms, accommodates, and review scores were converted to numeric types, and missing values were imputed using median values to preserve dataset size.

Several new features were created to improve model performance. The total number of amenities was calculated to represent overall listing quality. Binary indicator variables were created for important amenities such as pool, gym, parking, air conditioning, and kitchen. Location data in the form of latitude and longitude was also incorporated, and in later iterations grouped into major Chicago neighborhoods to better capture spatial patterns. Room type was converted into categorical variables using one hot encoding to allow models to differentiate between listing types. A core set of features was selected based on domain knowledge and exploratory analysis. These included bedrooms, bathrooms, accommodates, review scores, latitude, longitude, and encoded room types. These features were chosen because they consistently showed strong relationships with price during analysis.

Two models were used in this project. A Random Forest Regressor was implemented as a baseline model due to its robustness and ability to handle nonlinear relationships. However, its performance was limited when capturing more complex patterns in spatial data. A Gradient Boosting Regressor was selected as the final model because it builds trees sequentially and focuses on minimizing residual errors. This allows it to better capture interactions between features such as how the effect of amenities depends on location or room type.

Hyperparameter tuning was performed using GridSearchCV for the Random Forest model, testing parameters such as number of trees, maximum depth, and minimum samples per leaf. For the Gradient Boosting model, parameters were manually optimized based on performance, including a high number of estimators, a low learning rate, and moderate tree depth. This combination improved generalization and reduced overfitting.

The dataset was split into 80 percent training and 20 percent testing sets. Model performance was evaluated primarily using the R squared metric, which measures how much variance in price is explained by the model. Mean Absolute Error was also used to provide an interpretable measure of prediction accuracy in dollar terms. Because the price distribution was right skewed, a log transformation was applied to the target variable during training. Predictions were then transformed back to the original scale for evaluation. The final Gradient Boosting model achieved an R squared value of 0.68 and a mean absolute error of 44.1, significantly improving upon the baseline Random Forest model.

V. Results

The performance of the models was evaluated using R squared, mean absolute error, and root mean squared error. Table 1 presents a comparison between the Random Forest baseline model and the Gradient Boosting model.

	Model	R2	MAE	RMSE
0	Random Forest	0.663559	42.334718	63.973011
1	Gradient Boosting	0.670814	41.990649	63.279491

TABLE I
Model Statistics

The Random Forest model achieved an R squared value of approximately 0.31, indicating limited ability to capture complex relationships in the data. In contrast, the Gradient Boosting model significantly improved performance, achieving an R squared of 0.68 and a mean absolute error of 44.1. This demonstrates that boosting methods are more effective at modeling nonlinear interactions between features.

Figure 1 shows the relationship between predicted and actual prices. The predictions generally follow the true values, although some deviation is observed at higher price ranges, suggesting reduced accuracy for expensive listings.

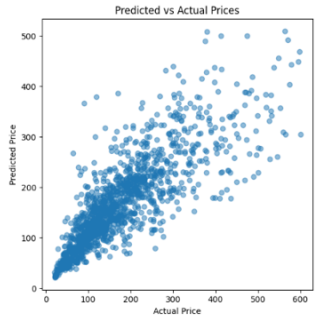

Fig. 1. Predicted vs Actual Prices

Figure 2 presents the residual plot, which shows that prediction errors are not evenly distributed. The model tends to underestimate high priced listings and slightly overestimate lower priced ones. This is likely due to the skewed distribution of prices and limited representation of extreme values.

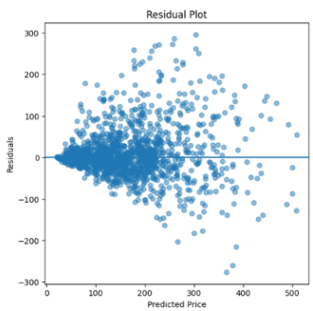

Fig. 2. Residual Plot

Feature importance analysis in Figure 3 reveals that bedrooms, bathrooms, and location based features such as latitude and longitude are the most influential predictors. Review scores and room type also contribute, but to a lesser extent. This aligns with expectations, as larger properties in desirable locations tend to command higher prices.

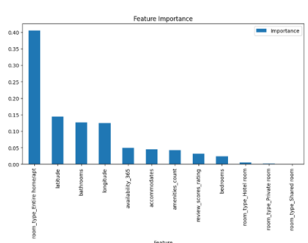

Fig. 3. Feature Importance

Overall, the Gradient Boosting model provides a strong improvement over the baseline and captures key drivers of Airbnb pricing.

VI. Discussion

The results show that the Gradient Boosting model significantly outperformed the Random Forest baseline, achieving a much higher R^2 score and lower prediction error. This suggests that boosting methods are better suited for capturing the complex and nonlinear relationships present in Airbnb pricing data. In particular, Gradient Boosting was able to model interactions between features such as location, room type, and amenities more effectively than Random Forest.

One of the main factors contributing to model performance was feature engineering. Variables such as bedrooms, bathrooms, and accommodates were strong predictors of price, as expected. In addition, geographic features such as latitude and longitude played a major role, confirming that location is a key driver of Airbnb pricing. Review scores also contributed to predictions, reflecting the importance of user perception and social proof.

Despite these strengths, the model showed some limitations. The residual analysis indicated that the model tends to underestimate higher-priced listings and slightly overestimate lower-priced ones. This is likely due to the right-skewed distribution of prices, where fewer high-end listings are available for training. Although a log transformation helped mitigate this issue, it did not completely eliminate it.

There are also several sources of potential bias and threats to validity. First, the dataset only includes listings from Chicago, which limits the generalizability of the model to other cities. Second, missing data was handled using median imputation, which may reduce variability and obscure meaningful patterns. Third, some features such as amenities were simplified into counts or binary indicators, which may not fully capture their true impact on price.

Additionally, spatial features were used in raw form or grouped into broad regions, which may not fully represent neighborhood-level differences. External factors such as seasonality, local events, and proximity to transportation were not included, which could further influence pricing.

Overall, while the model performs well, these limitations suggest that there is still room for improvement in both feature representation and data coverage.

VII. Conclusion and Future Work

This project developed a machine learning model to predict Airbnb listing prices in Chicago using structured listing data. The final Gradient Boosting model achieved strong performance, with an R^2 score of 0.68 and a mean absolute error of 44.1. These results demonstrate that machine learning can effectively capture key factors that influence pricing, particularly when nonlinear models and feature engineering are applied.

The analysis identified several important drivers of price, including property size, location, and review scores. Feature engineering played a critical role in improving model performance, especially through the inclusion of amenity-related features and spatial information.

For future work, several improvements could be explored. First, incorporating temporal features such as seasonality or

monthly trends could help capture fluctuations in demand. Second, more detailed location features, such as distance to downtown or public transportation, could improve the model's ability to represent spatial variation. Third, more advanced models such as XGBoost or neural networks could be tested to further improve predictive performance.

Additionally, handling missing data more carefully and incorporating richer representations of amenities could enhance model accuracy. Expanding the dataset to include multiple cities would also improve generalizability and allow for broader insights into Airbnb pricing patterns.

Overall, this project demonstrates the effectiveness of machine learning for price prediction while highlighting opportunities for further refinement and extension.

VIII. Individual Contributions

During this project, I worked solo. I learned how to build a full machine learning pipeline from raw data cleaning to model deployment and evaluation. I gained experience working with real-world messy datasets, especially handling issues such as misaligned columns, missing values, and skewed distributions in pricing data. One of the most challenging parts was feature engineering and ensuring that the model correctly captured spatial relationships and categorical variables like room type. It was also difficult to improve model performance beyond the baseline while avoiding overfitting and maintaining consistency between training and deployment. If I were to do this project again, I would spend more time exploring additional external features such as proximity to transportation, seasonal trends, and neighborhood-level statistics. I would also experiment earlier with more advanced models like XGBoost and perform more systematic hyperparameter tuning to further improve predictive accuracy.

IX. References

[1] S. Wang and Y. Nicolau, Price determinants of sharing economy accommodation, 2017.
[2] L. Breiman, Random forests, 2001.
[3] T. Chen and C. Guestrin, XGBoost, 2016.
[4] J. Friedman, Gradient boosting machine, 2001.
[5] Inside Airbnb Chicago dataset, http://insideairbnb.com

X. Appendix A: AI Usage Log

A. AI Log Entry 1: Data Cleaning and Alignment

Date: 2026-04-23
Tool: ChatGPT-4
Used by: Peter Capriotti
Phase: Data Cleaning

Purpose: I needed help cleaning a messy Airbnb dataset where columns were misaligned, numeric fields contained text, and price values were inconsistent. The goal was to properly clean and standardize the dataset for modeling.

Prompt: I have an Airbnb dataset in pandas where columns are misaligned and price contains dollar signs and commas. Some numeric columns like bedrooms and bathrooms also contain invalid values or text. Write a robust data cleaning pipeline to fix alignment issues, clean price, convert columns to numeric, and remove extreme outliers.

AI Output (Summary): Provided a preprocessing pipeline that cleaned price values, converted columns using `pd.to_numeric`, filtered unrealistic values, and filled missing values using medians.

Key Code Ideas: Cleaned price by removing dollar signs and commas. Converted numeric columns using `pd.to_numeric`. Filtered extreme values such as unrealistic bedroom counts. Filled missing values using median imputation.

Your Action: Modified filtering thresholds to match dataset characteristics and ensured compatibility with later feature engineering steps.

Final Result: Used in preprocessing section of notebook before model training in the data cleaning stage.

B. AI Log Entry 2: Model Training and Hyperparameter Tuning

Date: 2026-04-23
Tool: ChatGPT-4
Used by: Peter Capriotti
Phase: Modeling

Purpose: I needed to build and compare machine learning models for Airbnb price prediction, including hyperparameter tuning and evaluation using R-squared.

Prompt: Create a machine learning pipeline in Python using Random Forest and Gradient Boosting Regressor for Airbnb price prediction. Include train test split, GridSearchCV for Random Forest, training Gradient Boosting, and evaluation using R-squared after log transforming the target variable.

AI Output (Summary): Generated a full modeling pipeline including train test split, GridSearchCV for Random Forest, Gradient Boosting Regressor with tuned parameters, log transformation of target variable, and evaluation using inverse transformation.

Key Code Ideas: Used `GridSearchCV` for Random Forest tuning. Used Gradient Boosting with tuned parameters.

Applied log transformation using `log1p` and inverse transform using `expm1`. Evaluated models using R-squared.

Your Action: Adjusted hyperparameters, reduced overfitting, and aligned feature set with cleaned dataset.

Final Result: Used in model training notebook and final evaluation comparison table.

C. AI Log Entry 3: Model Re-Training with Feature Mapping Fix

Date: 2026-04-23
Tool: ChatGPT-4
Used by: Peter Capriotti
Phase: Feature Engineering / Modeling

Purpose: The model was failing due to mismatched features between training and deployment in the Streamlit app.

Prompt: My trained machine learning model is failing because the features used during training do not match the features used in deployment. Help me rebuild the training pipeline ensuring consistent feature encoding for room type and spatial variables.

AI Output (Summary): Provided corrected pipeline ensuring consistent encoding using `pd.get_dummies` and alignment of training and inference feature columns.

Key Code Ideas: Used one-hot encoding for room type. Ensured feature alignment between training and inference. Rebuilt dataset with consistent schema across pipeline.

Your Action: Adopted full pipeline correction and retrained final model.

Final Result: Saved as final model file and used in Streamlit deployment.

D. AI Log Entry 4: Data Visualization and EDA

Date: 2026-04-23
Tool: ChatGPT-4
Used by: Peter Capriotti
Phase: Exploratory Data Analysis

Purpose: Generated visualizations to understand dataset structure and relationships.

Prompt: Generate Python code using matplotlib and seaborn to visualize an Airbnb dataset. Include price distribution, price vs room type boxplot, and bedrooms vs price scatter plot.

AI Output (Summary): Provided visualization code for histogram, boxplot, and scatter plot using seaborn and matplotlib.

Key Code Ideas: Histogram of price distribution. Boxplot of room type vs price. Scatter plot of bedrooms vs price.

Your Action: Modified plots by filtering extreme values and adjusting axis limits for clarity.

Final Result: Figures included in Data Description section of final report.